Bird Classification based on Bird Sounds

Girish Rane
Department of Electronics and Telecommunication Engineering
College of Engineering, Pune
Pune, India
ranegl19.extc@coep.ac.in

Priti P. Rege
Department of Electronics and Telecommunication Engineering
College of Engineering, Pune
Pune, India
ppr.extc@coep.ac.in

Rashmika Patole
Department of Electronics and Telecommunication Engineering
College of Engineering, Pune
Pune, India
rkp.extc@coep.ac.in

***Abstract*— Automatic classification of bird calls plays an essential role in monitoring and protecting biodiversity. Classifying bird species is a challenging problem due to variant species present in the world. In this paper, we present the performance of Mel Frequency Cepstral Coefficients and Linear Frequency Cepstral Coefficients as features for classification of bird species. Exhaustive experimentation using Support Vector Machine, Artificial Neural Network and Convolution Neural Network used for classification is also reported. An accuracy of 93.37% is achieved with Convolution Neural Network and using a set of 39 MFCC features.**

***Keywords*— MFCC, LFCC, Deep Convolutional Neural Network(CNN), SVM**

I. INTRODUCTION

Acoustic communication is the primary mode of communication among birds. Birds are difficult to be traced, and therefore, their sounds are very effective in detecting birds by humans. Challenges of discriminative region localization make recognition of bird species difficult. It is difficult to identify bird sounds without blare.

The significant studies in bird recognition are based on visual inspection of the birds and require a trained observer. Each bird species contain more than one syllabus. Recognition of bird species involves feature extraction and classification. The extraction of features and choice of the classification algorithm affects the performance of the classification system. A variety of methods have been used for the automatic recognition of bird species. The deep learning model has recently attracted a lot of attention due to its high performance in building automated bird sound classification systems.

II. LITERATURE SURVEY

Authors in [1], have used MFCC for each bird species as features. The Gaussian Mixture Model(GMM) and Vector Quantization(VQ) models are used for identifying the most appropriate GMM component and numbered the clusters of VQ. In[2], Dynamic Time Warping (DTW) is utilized to differentiate the spectrograms of birds sound.

For identification of various bird species, different features like sets of Mel Frequency Cepstral Coefficients (MFCC) and Gammatone Frequency Cepstral Coefficients(GFCC) are considered as the features in [3]. Support Vector Machine(SVM), k-nearest neighbors (KNN), Artificial Neural Network(ANN) are used as classifiers. Various techniques employed in classification of speech are explored for the recognition of bird sounds. The researchers also used Hidden Markov Models(HMM) to classify data with the minimum number of training and testing samples[4].

In [5], six different species were compared with three different types of feature vectors. The classification algorithms used are k-nearest neighbors (KNN) and Support Vector Machine(SVM).In this paper, two kinds of feature extraction techniques based on MFCC and LFCC are proposed with Artificial Neural Network(ANN) and Support Vector Machine (SVM) and Convolution Neural Network (CNN) as classifiers. A convolution neural network is an efficient way to extract information and classification [6]. Jürgen T. Geiger, Björn Schuller, Gerhard Rigoll [10] used recordings of 10 acoustic environments and a sliding window approach and SVM to classify them. Authors in [13] have created a vector of 13 MFCC features and the first three Principal components. Further, they have used Vector Quantization to compress the vector. KNN is used as a classifier. P. Jancovic and M. Köküe [14] used Deep Neural Network-Hidden Markov Models(DNN-HMM) to recognize bird species.

III. METHODOLOGY

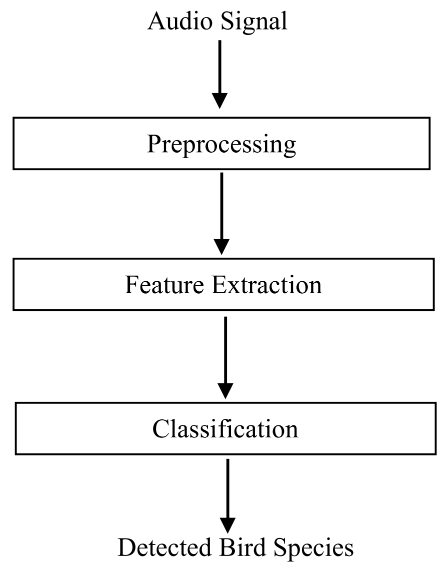


Figure 1: The block diagram of the proposed bird species recognition system

As shown in Figure 1, we have three parent blocks for the bird species recognition system. The blocks are discussed in the following subsections.

A) PREPROCESSING:

In the proposed studies, all bird sound recording files are taken from xeno-canto.com[7]. This website is dedicated to the bird's files dataset. Table 1 contains the list of species with scientific names and the number of files considered for the bird's sound classification. Each audio file is divided into 3 seconds chunk files. Each chunk must contain at least one syllabus of bird species. As the recordings have background noise, the subtraction algorithm [8] is used for noise removal. The sampling frequency is 16000Hz. For framing, we have used a 20 milliseconds window with 50% overlap, and the size of FFT is 512. The overall dataset is split into the training dataset with 70% samples and the testing dataset with the remaining 30 % data.

Table 1 Dataset used

Sr. No.	**Bird species**	**Scientific name**	**Number of Files**
1	Red-whiskered Bulbul(RB)	*Pycnonotus jocosus*	236
2	Great Tilt(GT)	*Parus major major*	243
3	Willow Warbler(WW)	*Phylloscopus trochilus*	262
4	European Pied Flycatcher(EF)	*Ficedula hypoleuca*	264
5	Common Blackbird(CB)	*Turdus merula*	256
6	Common Redstart(CR)	*Phoenicurus phoenicurus*	188
7	Greenish Warbler(GW)	*Phylloscopus torchiloides viridanus*	207
8	Blue Jay(BJ)	*Cyanocitta cristata*	276
9	Common Snipe(CS)	*Gallinago gallinago*	244
10	Whistling Duck(WD)	*Dendrocygna viduata*	182
			Total2358

B) FEATURE EXTRACTION

MFCC and LFCC are usually used for speech recognition. Each speech signal is pre-emphasized and transformed into the frequency domain. The frequency-domain signal is divided into small frames. The normal frequency is converted to Mel scale frequency. The conversion formula is given by,

$$\text{Mel(f)} = S_k = 2595 * log(1 + \frac{f}{700}) \qquad (1)$$

The spectrum and the Mel scale filter bank are multiplied. Following this, each filter bank's log energy is computed. The Cepstral coefficients are derived using discrete cosine transform. The first thirteen coefficients, delta coefficients (Δ) and double delta coefficients ($\Delta\Delta$), are selected for bird classification.

The main difference between LFCC and MFCC is in the selection of filter banks. LFCC uses a Linear filter bank, while MFCC uses a Mel filter Bank. The use of a linear frequency filter bank is to detect sounds at a higher frequency. The linear frequency cepstral coefficients (LFCC) are calculated as,

$$C_{in}[n,m] = \frac{1}{R}\sum_{l=0}^{R-1}\left(\log\{E_{in}(n,l)\}\cos(\frac{2\pi}{R}lm)\right) \qquad (2)$$

Where R is the number of filters, and $E_{in}(n,l)$ represents the energy of each frame.

C) CLASSIFICATION

For classification, we have used SVM, ANN, and CNN. In the feature extraction method, the feature vector of each frame is calculated, and each feature vector contains delta coefficient(Δ) and double delta coefficients.

1) SUPPORT VECTOR MACHINE(SVM)

Support Vector Machines are based on structural risk minimization [13]. SVM creates a hyperplane that separates features from two or more classes. The distance between hyperplane and support vector is called marginal distance. The aim is to maximize this marginal distance and minimize training error. It uses kernel version and accurate for classification and rapid learning. The selection of kernel is based on application and performance. The linear kernel used for high-dimensional features. For the SVM classifier, the mean value of each frame is used for the classification of different species. The SVM classifier uses a training set of labels and learns classification functions automatically. Each species represent one class.

2) ARTIFICIAL NEURAL NETWORK(ANN)

ANN models consist of multiple layers viz. the input layer, hidden layer followed by an output layer. Each layer contains nodes, also called neurons, that process data. The accuracy of a neural network depends on the number of hidden layers. In the input layer, set of neuron represents the coefficients, and the output layer represents the number of species .In the ANN network, each hidden layer learns the pattern of the previously selected hidden layer. The hidden layer contains three fully connected layers. Each fully connected layer identifies features from the previous layer and correlates them. ReLU activation function gives the best performance of each hidden layer. At the output layer, the Softmax activation function predicts the probability of each class. A total of 10 neurons representing ten bird classes are used in the output layer. The highest probability out of 10 classes defines the output of the input layer.

3) Convolution Neural Network(CNN)

For Convolution Neural Network model has two convolution layers followed by two dense layers. Each convolution layer contains a filter size of 3x3 with stride 1.. For training data, we used categorical cross-entropy with Adam optimizer. The learning rate is 0.001. The input value of convolution neural network MxN, where M represents the number of frames and N is the number of features present in each bird file. The model is trained and tested on 13, 26, and 39 coefficients.

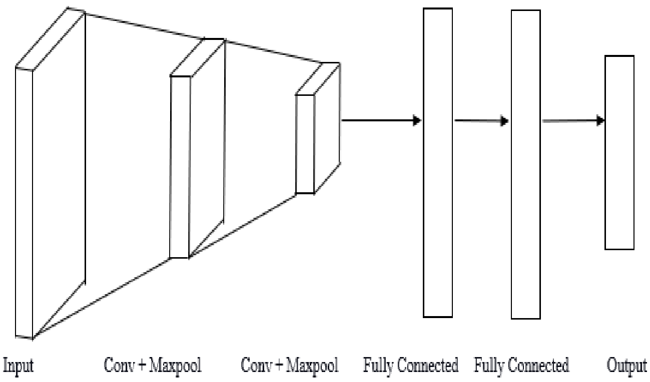


Figure 2 CNN Architecture

As shown in Figure 2, each convolution layer learns a pattern of data and reduces the dimensions. The first two convolution layers contain 32 and 64 kernel filters, respectively. The number of filters always equals the number of feature maps of the next layer. The filter size of 3x3 with the Max-pool layer brings down the size of the feature map. After two convolution layers, flatten layer is used. Flatten layer converts 2-Dimensional layer to a 1-Dimensional array, also called a fully connected layer. Every neuron from the fully connected layer is coupled with another fully connected layer.

Each fully connected layer uses ReLU as activation function. We have trained the model for 30 epochs. Categorical cross-entropy is used as a loss function. This loss function summarizes the average value of actual and predicted classes. At the output layer, the Softmax activation function calculates the probability of each class.

IV. RESULT AND DISCUSSION

Table 1 depicts the description of species. There are 2358 individual bird files from 10 species. The Species Recognition Accuracy (SRA) is calculated as [8]:

$$SRA = \frac{CS_A}{NS_A}\ x\ 100 \qquad (3)$$

CS_A represents the number of correctly classified files representing various bird species. The total number of test files used are NS_A. The Reliability in expressing the species is given by:

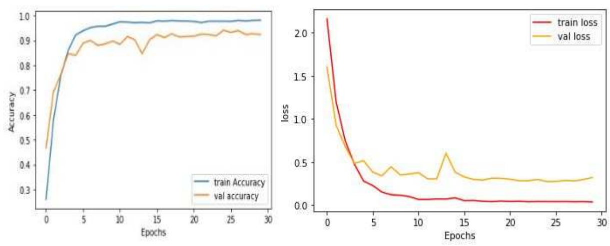
$$Rel = \frac{CS_A}{MS_A}\ x\ 100 \qquad (4)$$

Here, CS_A represents the number of species that are correctly classified. The total number of test files is MS_A. We calculate the Accuracy of the model as :

$$O.A. = \frac{C_A}{M_A}\ x\ 100 \qquad (5)$$

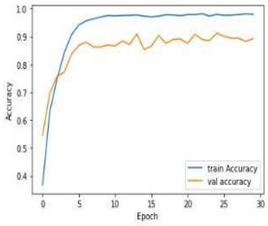
Where $M_{.A.}$ is the number of test files, and $C_{.A.}$ represents the correctly classified files.

The OA for MFCC is 82.37% for MFCC and 85.53% for LFCC in SVM. The confusion matrix is shown in Table 2 and Table 3. The maximum accuracy of ANN 91.60% for MFCC (13) features and 92.02% for LFCC (13) features. The OA accuracy of MFCC with 39 features is 93.37%, and LFCC with 39 features is 92.31%. The performance of the result highly depends on the removal of noise.

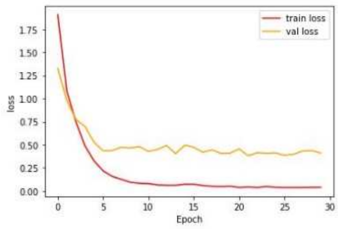


(a) (b)

Figure 3. (a) Accuracy curve (b) Loss curve of 39 MFCC Features



(a)



(b)

Figure 4. (a) Accuracy curve (b) Loss curve of 39 LFCC Features

Table 4 compares the performance of our scheme with the performance of the models in [9] and [13].As the number of coefficients used for classification increases, classification accuracy also increases. The selection of MFCC for classific ation gives better results compared to LFCC. As far as classi fiers are concerned, the best results are obtained with 39 MF CC coefficients and CNN as a classifier.

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Table 2: Confusion matrix for bird classification using (13) MFCCs

	Number of correctly recognized birds in the specific class										Acc(%)
	RB	GT	WW	EF	CB	CR	GW	BJ	CS	WD	
RB	**36**	1	2	0	0	0	0	1	4	0	**82**
GT	3	**45**	0	1	0	0	2	0	0	0	**88**
WW	2	0	**37**	4	0	0	1	4	0	0	**77**
EF	0	0	9	**47**	1	0	0	1	0	0	**81**
CB	0	0	2	6	**43**	2	0	0	0	0	**81**
CR	1	0	5	0	0	**19**	0	0	0	0	**76**
GW	4	1	3	0	0	0	**48**	1	0	0	**84**
BJ	0	0	0	0	0	0	4	**43**	8	6	**70**
CS	1	0	0	0	0	0	0	0	**39**	1	**95**
WD	1	0	0	0	0	0	0	2	0	**36**	**92**
%Rel	**75**	**96**	**64**	**81**	**98**	**90**	**87**	**83**	**76**	**84**	
Overall Accuracy: 82.38 %											

Table 3: Recognition result with LFCC (13) features

	Number of correctly recognized birds in the specific class										Acc(%)
	RB	GT	WW	EF	CB	CR	GW	BJ	CS	WD	
RB	**50**	0	1	0	0	0	0	2	3	0	**89**
GT	1	**38**	0	1	2	0	0	0	0	0	**90**
WW	0	0	**37**	9	0	1	2	0	0	3	**71**
EF	0	2	1	**54**	0	0	0	0	0	0	**95**
CB	0	0	1	5	**42**	0	1	0	0	0	**86**
CR	0	0	3	2	0	**23**	0	2	0	0	**77**
GW	0	0	3	0	0	0	**57**	1	0	0	**93**
BJ	2	0	0	0	0	0	4	**39**	1	3	**80**
CS	4	0	0	0	0	0	0	8	**34**	0	**74**
WD	0	0	0	0	1	0	0	0	0	**34**	**97**
%Rel	**88**	**95**	**80**	**76**	**93**	**96**	**89**	**75**	**89**	**85**	
Overall Accuracy: 85.53%											

Table 4. Comparison of classification accuracies for MFCC and LFCC

Paper	Feature /Classifier used	Accuracy
Bang et al. [13]	MFCC+ PCA/ KNN	82
Patole et al. [9]	MFCC /SVM MFCC/ ANN	91.7 91.6
Proposed technique	MFCC (13 coefficients)/ SVM LFCC (13 coefficients)/ SVM	89.10 85.87
	MFCC (13 coefficients)/ ANN LFCC (13 coefficients)/ ANN	91.60 92.02
	MFCC (13 coefficients)/ CNN LFCC (13 coefficients)/ CNN	89.68 89.27
	MFCC (26 coefficients)/ CNN LFCC (26 coefficients)/ CNN	92.55 91.50
	MFCC (39 coefficients)/ CNN LFCC (39 coefficients)/ CNN	93.37 92.31

V. CONCLUSION AND FUTURE WORK

In this paper, three sets of classifiers and two feature sets are selected to classify bird species. CNN with 39 feature coefficients gives the highest accuracy with both feature sets. An accuracy of 93.37 % has been achieved. The accuracy can be improved by selecting the number of features in the feature set. The performance of the model can be further enhanced by suppressing background noise in bird sound recordings.

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